Wireless Networking:-

Tasks:-

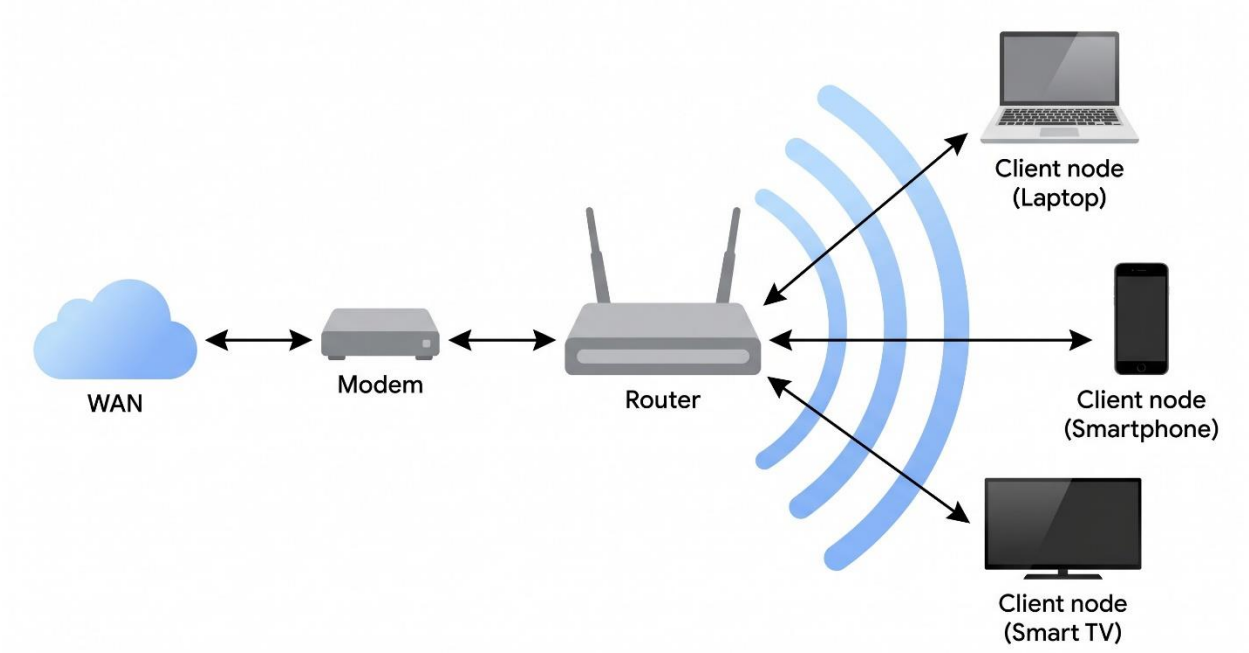
1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Ans.:-

Wi-Fi Standard	Wi-Fi Name	Frequency Band	Maximum Speed	Key Difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Up to 600 Mbps	First standard to support both 2.4 GHz and 5 GHz using MIMO technology for improved speed and range.
802.11ac	Wi-Fi 5	5 GHz only	Up to 6.9 Gbps	Much faster than 802.11n, uses wider 80/160 MHz channels and MU-MIMO for better performance.
802.11ax	Wi-Fi 6	2.4 GHz and 5 GHz (6 GHz with Wi-Fi 6E)	Up to 9.6 Gbps	Uses OFDMA and improved MU-MIMO to provide higher speeds, lower latency, and better performance in crowded environments.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

Ans.:-



**3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.**

Ans.:-

Available Wi-Fi Networks :-

SSID (Wi-Fi Name)	Signal Strength	Security Type	Wi-Fi Standard	Recommendation
GTPLPG6	96%	WPA2-Personal	802.11n	Best Choice
Airtel_abhi_4769	64%	WPA2-Personal	802.11ax (Wi-Fi 6)	Good Choice
GTPL	70%	WPA2-Personal	802.11n	Good Choice
Bharat21	32%	WPA2-Personal	802.11n	Weak Signal
dnafi_4G	34%	WPA2-Personal	802.11n	Weak Signal
Jay 307	18%	WPA2-Personal	802.11n	Very Weak Signal

Which network would I choose and why ?

I would choose GTPLPG6 because:

- It has the strongest signal strength (96%), ensuring the fastest and most stable connection.
- It uses WPA2-Personal security, which provides strong encryption for a home Wi-Fi network.
- A strong signal generally offers better performance for web browsing, video streaming, online gaming, and video conferencing.

Although Airtel_abhi_4769 supports the newer 802.11ax (Wi-Fi 6) standard, its 64% signal strength is lower than GTPLPG6. Therefore, GTPLPG6 is the better overall choice in your current location.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

Ans.:-

SSID Hiding in Wi-Fi Networks:-

SSID (Service Set Identifier) is the name of a Wi-Fi network that appears when you search for available wireless networks on your phone, laptop, or tablet. SSID hiding is a router feature that stops broadcasting this network name, making it invisible in the list of available Wi-Fi networks. To connect to a hidden network, users must manually enter the SSID (network name) and password.

How SSID Hiding Works:-

Normally, a Wi-Fi router sends out beacon frames that include the network's SSID so nearby devices can detect it. When SSID hiding is enabled, the router still sends beacon frames, but the SSID field is left blank. The network still exists and functions normally, but users must know the exact network name to connect.

Advantages of Hiding the SSID :-

1. **Reduces Visibility:** The Wi-Fi network is not displayed in the list of nearby networks, making it less noticeable to casual users.
2. **Discourages Unauthorized Users:** People looking for easy-to-find networks may ignore a hidden network.
3. **Adds a Small Layer of Privacy:** Visitors or neighbors are less likely to notice your network name.

Disadvantages of Hiding the SSID:-

1. **Not Real Security:** Hidden SSIDs can still be discovered using Wi-Fi analysis or packet-sniffing tools, so it does not stop experienced attackers.
2. **Less Convenient:** Every new device must be configured manually by entering the exact SSID and password.
3. **Connection Issues:** Some devices may take longer to connect or may repeatedly search for the hidden network, which can slightly affect battery life.

Example:-

- **Visible Network:**
  SSID: **Home_WiFi** → Appears automatically in the Wi-Fi list.
- **Hidden Network:**
  SSID: **Home_WiFi** → Does not appear in the Wi-Fi list. Users must manually enter **Home_WiFi** and its password to connect.